

Narenkarthik Kesavamoorthy

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Summary

Final-year B.Tech CSE (IoT) student working on robot software end to end: STM32 firmware, MuJoCo and Gazebo simulation, reinforcement and imitation learning, and fault-tolerant multi-robot systems.

Education

Shiv Nadar University Chennai, B.Tech Computer Science and Engineering (IoT) 2023 – 2027 (expected)
CGPA: 8.42 / 10 **Coursework:** Operating Systems, Microprocessors, Computer Architecture, Digital Design

Experience

Research Intern, ROAR Lab, Singapore University of Technology and Design May 2026 – Jul 2026
Supervisor: Prof. Mohan Rajesh Elara Singapore

- Designed and implemented the failsafe mechanism in the secondary control system (Embedded C, STM32) of *Saber*, an all-terrain quadruped robot; now being integrated into the robot.
- Diagnosed a faulty motor by tracing its circuitry to the specific failed chip, and helped recover the robot overnight from an unexpected multi-motor failure.
- Benchmarked Open-RMF-type fleet-management middleware platforms into a verified vendor and feature database.
- Reviewed technical proposals for an AI Centre of Excellence initiative, incorporating stakeholder feedback.

AI/Automation Intern, Yakkay Technologies, Chennai May 2025 – Jun 2025

- Built an HR-automation prototype in Python on a pre-trained model to automate hr resume checking.

Current Work

Capstone: Reinforcement Learning for Generalized Door Manipulation 2026 – present
SolidWorks, MuJoCo, Python, Gymnasium, Stable-Baselines3 (SAC) Team of 3, mentor Dr. Priya GL [GitHub](#)

- Testing whether domain-randomized training helps an anthropomorphic hand open *unseen* doors, and whether separate reach, handle and door policies (6, 12, 4 actions) make high-DOF RL tractable.
- Designed a 200 mm, 18-DOF hand (2-DOF wrist, universal-joint thumb) and a spring-loaded door in SolidWorks; built a SolidWorks → sw2robot → MuJoCo pipeline with Python tools that fix joint axes, set anatomical zeros and size actuators from gravity loads; Gymnasium environment with SAC training in place.

Projects

Swarm-Fleet: Decentralized Multi-Robot Task Allocation 2026
ROS 2 Jazzy, Gazebo, Zenoh, Raft, Docker, Kubernetes [GitHub](#)

- Built a ROS 2 warehouse swarm where robots bid BFS path cost and agree on winners with no central dispatcher.
- Ran each robot as a Kubernetes pod bridged by a Zenoh router, with a 3-replica Raft task announcer; chaos tests showed ~7s leader failover with tasks still completing, and exposed documented failure modes.

Behavior Cloning and DAgger for Robot Navigation 2026
ROS 2, Gazebo Harmonic, PyTorch, TurtleBot3 [GitHub](#)

- Trained an MLP on 2D LiDAR and goal input to imitate a gap-following expert (~54k samples, episode-level splits); on 100 unseen layouts DAgger raised success from 64% to 73% and cut collisions from 23% to 7%.
- An expert-memory ablation ruled out one cause of the plateau, pointing to averaging of ambiguous steering.

Other projects

- **Smart-AMBU** (Arduino, ESP32, MQTT, Node-RED): live airway pressure and breathing rate for manual resuscitators, with high-pressure and mask-seal alarms. [GitHub](#)
- **GLOF early warning**, Smart India Hackathon 2024 (Google Earth Engine): glacier water and ice area from Sentinel-2 imagery for flood early warning. [GitHub](#)

Technical Skills

Languages & Systems: C, Embedded C, Python, Bash, Docker, Kubernetes, Linux, Git
Robotics & AI: ROS 2, Gazebo, MuJoCo, Gymnasium, PyTorch, RL, imitation learning
Embedded, IoT & Tools: STM32, ESP32, Arduino, MQTT, Node-RED, SolidWorks, OpenCV, Google Earth Engine

Awards & Certifications

Gold, Melbourne Design Awards 2026 (Better Future Awards): *Saber* team, SUTD ROAR Lab 2026
Google Cloud Career Launchpad: completed all courses and labs in the Cloud Engineer track Dec 2025

Leadership & Volunteering

Club Head, Omnia, SNU Chennai: campus club for animal welfare, environment and inclusion 2026-2027
Volunteer, Raikan Ilmu 2026, Singapore: presented *Saber* to visitors at the SUTD booth 2026